secretion of thyroid. It is characterised by enlargement of thyroid gland, increased BMR (50% - 100%), elevated respiratory and excretory rates, increased heart beat, high BP, increased body temperature, protrusion of eyeball and weakness of eye muscles and weight loss. (Figure 11.13)



Figure. 11.13 Grave's disease

Simple goitre is also known as **Endemic goitre**. It is caused due to hyposecretion of thyroxine. The symptoms includes enlargement of thyroid gland, fall in serum thyroxine level, increased TSH secretion. (Figure 11.14)



Figure. 11.14 Simple goitre

Tetany is caused due to the hyposecretion of parathyroid hormone (PTH). Due to hyposecretion of PTH serum calcium level decreases (Hypocalcemia), as a result serum phosphate level increases. Calcium and phosphate excretion level decreases. Generalized convulsion, locking of jaws increased heart beat rate, increased body temperature, muscular spasm are the major symptoms of tetany.

Hyperparathyroidism is caused due to excess PTH in blood. Demineralisation of bone, cyst formation, softening of bone, loss of muscle tone, general weakness, renal disorders are the symptoms of hyperparathyroidism.

Addison's disease is caused due to hyposecretion of glucocorticoids and mineralocorticoids from the adrenal cortex. Muscular weakness, low BP., loss of appetite, vomiting, hyper pigmentation of skin, low metabolic rate, subnormal temperature, reduced blood volume, weight loss are the symptoms that occur in Addison's disease (Figure 11.15). Reduced aldosterone secretion increases urinary excretion of Na Cl. and water and decreases potassium excretion leading to dehydration.

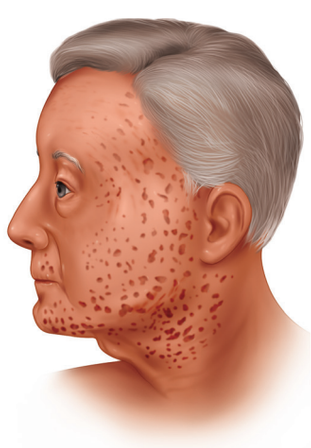


Figure. 11.15 Addison's disease

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